

Standard Model Signal in Cosmological Higgs Collider

1 Top Quark Signal

Higgs bulk-boundary propagator is a massless propagator:

$$G_{\pm}(k; \tau) = \frac{H^2}{2k^3} (1 \pm ik\tau) e^{\mp ik\tau} \quad (1)$$

The top quark kinetic term in the dS background is:

$$\mathcal{L}_{\text{Kin}} = \frac{\sqrt{-g}}{a(\tau)} (i\bar{t}_L \bar{\sigma}^\mu D_\mu t_L + i\bar{t}_R \bar{\sigma}^\mu D_\mu t_R) \quad (2)$$

where the covariant derivative is $D = \partial + \Gamma$.

The Yukawa coupling term is:

$$\mathcal{L}_{\text{Yuk}} = -\sqrt{-g} \frac{h_0 + \delta h}{\sqrt{2}} (y^t \bar{t}_L t_R + y^t \bar{t}_R t_L) \quad (3)$$

By redefinition, the Dirac fermion can be split to two decoupled 2 components Weyl spinors with Majorana mass $m = \frac{h_0 y^t}{\sqrt{2}}$:

$$\mathcal{L}_{\text{Kin}} + \mathcal{L}_{\text{Yuk}} = \mathcal{L}_0(\psi_1) + \mathcal{L}_0(\psi_2) \quad (4)$$

where a set of Majorana fermion theory is:

$$\mathcal{L}_0(\psi) = \frac{\sqrt{-g}}{a(\tau)} \left(i\psi^\dagger \bar{\sigma}^\mu D_\mu \psi - \frac{1}{2} m \psi \psi - \frac{1}{2} m \psi^\dagger \psi^\dagger \right) + \sqrt{-g} \frac{\delta h y^t}{\sqrt{2}} \left(-\frac{1}{2} \psi \psi - \frac{1}{2} \psi^\dagger \psi^\dagger \right) \quad (5)$$

The momentum direction is defined as:

$$\hat{\mathbf{k}} = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta) \quad (6)$$

And the corresponding helicity eigenspinors are:

$$h_+(\theta, \varphi) = \begin{pmatrix} e^{-i\frac{\varphi}{2}} \cos \frac{\theta}{2} \\ e^{i\frac{\varphi}{2}} \sin \frac{\theta}{2} \end{pmatrix}, \quad h_-(\theta, \varphi) = \begin{pmatrix} -e^{-i\frac{\varphi}{2}} \sin \frac{\theta}{2} \\ e^{i\frac{\varphi}{2}} \cos \frac{\theta}{2} \end{pmatrix} \quad (7)$$

After the redefinition $\bar{\psi} = a^{3/2}(\tau) \psi$, the vertex become:

$$\mathcal{L}_{\text{int}} = -a(\tau) \frac{1}{2} \frac{\delta h y^t}{\sqrt{2}} (\bar{\psi} \bar{\psi} + \bar{\psi}^\dagger \bar{\psi}^\dagger) \quad (8)$$

The mode function are:

$$u_+(\tau, k) = \frac{e^{-i\pi/4}\tilde{\nu}e^{\pi\tilde{\nu}/2}\Gamma(-i2\tilde{\nu})}{\Gamma(1-i\tilde{\nu})}(-2k\tau)^{i\tilde{\nu}}e^{-ik\tau}{}_1F_1(1+i\tilde{\nu}; 1+i2\tilde{\nu}; i2k\tau) + (\tilde{\nu} \rightarrow -\tilde{\nu}), \quad (9)$$

$$u_-(\tau, k) = \frac{e^{-i\pi/4}\tilde{\nu}e^{\pi\tilde{\nu}/2}\Gamma(-i2\tilde{\nu})}{\Gamma(-i\tilde{\nu})}(-2k\tau)^{i\tilde{\nu}}e^{-ik\tau}{}_1F_1(+i\tilde{\nu}; 1+i2\tilde{\nu}; i2k\tau) + (\tilde{\nu} \rightarrow -\tilde{\nu}), \quad (10)$$

$$v_+(\tau, k) = \frac{e^{-i\pi/4}\tilde{\nu}e^{\pi\tilde{\nu}/2}\Gamma(-i2\tilde{\nu})}{\Gamma(-i\tilde{\nu})}(-2k\tau)^{i\tilde{\nu}}e^{-ik\tau}{}_1F_1(+i\tilde{\nu}; 1+i2\tilde{\nu}; i2k\tau) + (\tilde{\nu} \rightarrow -\tilde{\nu}), \quad (11)$$

$$v_-(\tau, k) = \frac{e^{-i\pi/4}\tilde{\nu}e^{\pi\tilde{\nu}/2}\Gamma(-i2\tilde{\nu})}{\Gamma(1-i\tilde{\nu})}(-2k\tau)^{i\tilde{\nu}}e^{-ik\tau}{}_1F_1(1+i\tilde{\nu}; 1+i2\tilde{\nu}; i2k\tau) + (\tilde{\nu} \rightarrow -\tilde{\nu}). \quad (12)$$

and we have:

$$u_+ = iu_- = iv_+ = v_- \quad (13)$$

So the 2 components Weyl spinor greater and smaller propagators of Type- $\langle\bar{\psi}\bar{\psi}^\dagger\rangle$ are: [1, 2]:

$$\begin{aligned} D_{>\alpha\dot{\beta}}(\mathbf{k}; \tau_1, \tau_2) &= \sum_s \xi_{s,\alpha}(\tau_1, \mathbf{k}) \xi_{s,\dot{\beta}}^\dagger(\tau_2, \mathbf{k}) = \sum_s u_s(\tau_1, \mathbf{k}) u_s^\dagger(\tau_2, \mathbf{k}) h_{s,\alpha}(\mathbf{k}) h_{s,\dot{\beta}}^\dagger(\mathbf{k}), \\ D_{<\alpha\dot{\beta}}(\mathbf{k}; \tau_1, \tau_2) &= \sum_s -\chi_{s,\dot{\beta}}^\dagger(\tau_2, \mathbf{k}) \chi_{s,\alpha}(\tau_1, \mathbf{k}) = \sum_s -v_s(\tau_2, \mathbf{k}) v_s^\dagger(\tau_1, \mathbf{k}) h_{-s,\dot{\beta}}^\dagger(\mathbf{k}) h_{-s,\alpha}(\mathbf{k}) \end{aligned} \quad (14)$$

and the Type- $\langle\bar{\psi}\bar{\psi}\rangle$ are:

$$\begin{aligned} \hat{D}_{>\alpha\beta}(\mathbf{k}; \tau_1, \tau_2) &= \sum_s \xi_{s,\alpha}(\tau_1, \mathbf{k}) \chi_{s,\beta}(\tau_2, \mathbf{k}) = \sum_s s u_s(\tau_1, \mathbf{k}) v_s^\dagger(\tau_2, \mathbf{k}) h_{s,\alpha}(\mathbf{k}) h_{-s,\beta}(\mathbf{k}), \\ \hat{D}_{<\alpha\beta}(\mathbf{k}; \tau_1, \tau_2) &= \sum_s -\xi_{s,\beta}(\tau_2, \mathbf{k}) \chi_{s,\alpha}(\tau_1, \mathbf{k}) = \sum_s -s u_s(\tau_2, \mathbf{k}) v_s^\dagger(\tau_1, \mathbf{k}) h_{s,\beta}(\mathbf{k}) h_{-s,\alpha}(\mathbf{k}) \end{aligned} \quad (15)$$

and Type- $\langle\bar{\psi}^\dagger\bar{\psi}^\dagger\rangle$ are

$$\check{D}_{>\dot{\alpha}\dot{\beta}}(\mathbf{k}; \tau_1, \tau_2) = \sum_s \chi_{s,\dot{\alpha}}^\dagger(\tau_1, \mathbf{k}) \xi_{s,\dot{\beta}}^\dagger(\tau_2, \mathbf{k}) = \sum_s s v_s(\tau_1, \mathbf{k}) u_s^\dagger(\tau_2, \mathbf{k}) h_{-s,\dot{\alpha}}^\dagger(\mathbf{k}) h_{s,\dot{\beta}}^\dagger(\mathbf{k}), \quad (16)$$

$$\check{D}_{<\dot{\alpha}\dot{\beta}}(\mathbf{k}; \tau_1, \tau_2) = \sum_s -\chi_{s,\dot{\beta}}^\dagger(\tau_2, \mathbf{k}) \xi_{s,\dot{\alpha}}^\dagger(\tau_1, \mathbf{k}) = \sum_s -s v_s(\tau_2, \mathbf{k}) u_s^\dagger(\tau_1, \mathbf{k}) h_{-s,\dot{\beta}}^\dagger(\mathbf{k}) h_{s,\dot{\alpha}}^\dagger(\mathbf{k}) \quad (17)$$

Specifically, the nonlocal parts of symmetric propagators are:

$$\left[D_{\alpha\dot{\beta}}^{(S)}(\mathbf{k}; \tau_1, \tau_2) \right]_{(\text{NLoc})} = 0 \quad (18)$$

$$\left[\hat{D}_{\alpha}^{(S)\beta}(\mathbf{k}; \tau_1, \tau_2) \right]_{(\text{NLoc})} = \sum_{f=\pm} D_f^{(S)}(\mathbf{k}; \tau_1, \tau_2) \delta_{\beta}^{\alpha} \quad (19)$$

$$\left[\check{D}_{\dot{\beta}}^{(S)\dot{\alpha}}(\mathbf{k}; \tau_1, \tau_2) \right]_{(\text{NLoc})} = \sum_{f=\pm} D_f^{(S)}(\mathbf{k}; \tau_1, \tau_2) \delta_{\dot{\beta}}^{\dot{\alpha}} \quad (20)$$

$$D_f^{(S)}(\mathbf{k}; \tau_1, \tau_2) = \left[\frac{|\tilde{\nu}| [\Gamma(-2if\tilde{\nu})]^2}{\Gamma(1-if\tilde{\nu})\Gamma(-if\tilde{\nu})} (4k^2\tau_1\tau_2)^{if\tilde{\nu}} \right] \quad (21)$$

For the 4pt box diagram, there are $2^4 = 16$ graphs contribute the nonlocal signal, which contain different choices of each vertex over $\delta h \bar{\psi} \bar{\psi}$ or $\delta h \bar{\psi}^\dagger \bar{\psi}^\dagger$.

But given that only $a = b = c = d = f$ terms will dominate, and $\left[D_{\alpha\beta}^{(S)}(\mathbf{k}; \tau_1, \tau_2) \right]_{\text{NLoc}} = 0$, there are only two types of diagram that contribute the leading nonlocal signal.

1.1 $4 - \delta h \bar{\psi} \bar{\psi}$ Diagram

The trispectrum by $4 - \delta h \bar{\psi} \bar{\psi}$ diagram:

$$\begin{aligned} \langle \delta h \rangle^{40} &= \left(\frac{y^t}{\sqrt{2}} \right)^4 \int \frac{d\tau_1}{(-H\tau_1)} \frac{d\tau_2}{(-H\tau_2)} \frac{d\tau_3}{(-H\tau_3)} \frac{d\tau_4}{(-H\tau_4)} \int \frac{d^3\mathbf{q}}{(2\pi)^3} \sum_{a,b,c,d=\pm} (abcd) \\ &\quad \times G_a(k_1, \tau_1) G_b(k_2, \tau_2) G_c(k_3, \tau_3) G_d(k_4, \tau_4) \\ &\quad \times \hat{D}_{ab\alpha}^\beta(\mathbf{q} + \mathbf{k}_1, \tau_1, \tau_2) \hat{D}_{bc\beta}^\gamma(\mathbf{q} + \mathbf{k}_s, \tau_2, \tau_3) \hat{D}_{cd\gamma}^\delta(\mathbf{q} + \mathbf{k}_3, \tau_3, \tau_4) \hat{D}_{da\delta}^\alpha(\mathbf{q}, \tau_4, \tau_1) \end{aligned} \quad (22)$$

Firstly we deal with the loop integral:

$$\tilde{B} = \int \frac{d^3\mathbf{q}}{(2\pi)^3} \hat{D}_{ff\alpha}^\beta(\mathbf{q} + \mathbf{k}_1, \tau_1, \tau_2) \hat{D}_{bc\beta}^\gamma(\mathbf{q} + \mathbf{k}_s, \tau_2, \tau_3) \hat{D}_{cd\gamma}^\delta(\mathbf{q} + \mathbf{k}_3, \tau_3, \tau_4) \hat{D}_{da\delta}^\alpha(\mathbf{q}, \tau_4, \tau_1) \quad (23)$$

The dominated nonlocal signal will come from soft internal line:

$$\begin{aligned} \tilde{B} &= \hat{D}_{ab\alpha}^\beta(\mathbf{k}_1, \tau_1, \tau_2) \hat{D}_{cd\gamma}^\delta(\mathbf{k}_3, \tau_3, \tau_4) \\ &\quad \times \int \frac{d^3\mathbf{q}}{(2\pi)^3} \left[\hat{D}_{bc\beta}^\gamma(\mathbf{q} + \mathbf{k}_s, \tau_2, \tau_3) \right]_{\text{NLoc}} \left[\hat{D}_{da\delta}^\alpha(\mathbf{q}, \tau_4, \tau_1) \right]_{\text{NLoc}} \end{aligned} \quad (24)$$

We can substitute the expression of soft internal lines, and will find only $a = b = c = d = f$ terms dominate in the left and right subgraph. So we just calculate these terms:

$$\begin{aligned} \tilde{B} &= \sum_{f=\pm} \hat{D}_{ff\alpha}^\beta(\mathbf{k}_1, \tau_1, \tau_2) \hat{D}_{ff\gamma}^\delta(\mathbf{k}_3, \tau_3, \tau_4) \\ &\quad \times \int \frac{d^3\mathbf{q}}{(2\pi)^3} D_f^{(S)}(\mathbf{q} + \mathbf{k}_s; \tau_2, \tau_3) \delta_\beta^\gamma D_f^{(S)}(\mathbf{q}; \tau_3, \tau_4) \delta_\delta^\alpha \\ &= \sum_{f=\pm} \hat{D}_{ff\alpha}^\beta(\mathbf{k}_1, \tau_1, \tau_2) \hat{D}_{ff\gamma}^\delta(\mathbf{k}_3, \tau_3, \tau_4) (\tau_1 \tau_2 \tau_3 \tau_4)^{if\tilde{\nu}} \times \mathcal{B}_{f,\beta}^{\gamma\alpha} \end{aligned} \quad (25)$$

where the bubble contribution:

$$\mathcal{B}_{f,\beta}^{\gamma\alpha} = \frac{\tilde{\nu}^2 [\Gamma(-2if\tilde{\nu})]^4}{[\Gamma(1 - if\tilde{\nu})]^2 [\Gamma(-if\tilde{\nu})]^2} \times \frac{2^{i4f\tilde{\nu}} k_s^{3+i4f\tilde{\nu}} \Gamma(-\frac{3}{2} - 2if\tilde{\nu}) [\Gamma(\frac{3}{2} + if\tilde{\nu})]^2}{(4\pi)^{3/2} \Gamma(3 + 2if\tilde{\nu}) [\Gamma(-if\tilde{\nu})]^2} \cdot \delta_\beta^\gamma \delta_\delta^\alpha \quad (26)$$

$$= \mathcal{B}_f \cdot k_s^{3+i4f\tilde{\nu}} \cdot \delta_\beta^\gamma \delta_\delta^\alpha \quad (27)$$

Then we turn to left subgraph:

$$\mathcal{T}_{f\alpha\kappa}^{(L)} = \left(\frac{y^t}{\sqrt{2}} \right)^2 \int \frac{d\tau_1}{(-H\tau_1)} \frac{d\tau_2}{(-H\tau_2)} \times G_f(k_1, \tau_1) G_f(k_2, \tau_2) \times \hat{D}_{ff\alpha\kappa}(\mathbf{k}_1, \tau_1, \tau_2) \cdot (\tau_1 \tau_2)^{if\tilde{\nu}} \quad (28)$$

$$\begin{aligned} &= \left(\frac{y^t}{\sqrt{2}} \right)^2 \frac{H^2}{4k_1^3 k_2^3} \frac{1}{k_1^{if\tilde{\nu}} k_2^{if\tilde{\nu}}} \\ &\quad \times \left[\hat{I}_{ff\alpha\kappa}^{-1+if\tilde{\nu}, -1+if\tilde{\nu}} - if \hat{I}_{ff\alpha\kappa}^{if\tilde{\nu}, -1+if\tilde{\nu}} - if \hat{I}_{ff\alpha\kappa}^{-1+if\tilde{\nu}, if\tilde{\nu}} - \hat{I}_{ff\alpha\kappa}^{if\tilde{\nu}, if\tilde{\nu}} \right] \end{aligned} \quad (29)$$

The integral seed is defined as:

$$\hat{\mathcal{I}}_{\oplus\oplus\alpha\beta}^{p_1p_2} = \begin{pmatrix} 0 & S_1^{p_1p_2} + S_2^{p_2p_1} \\ -(S_1^{p_2p_1} + S_2^{p_1p_2}) & 0 \end{pmatrix}_{\alpha\beta},$$

$$\hat{\mathcal{I}}_{\ominus\ominus\alpha\beta}^{p_1p_2} = \begin{pmatrix} 0 & S_3^{p_1p_2} + S_4^{p_2p_1} \\ -(S_3^{p_2p_1} + S_4^{p_1p_2}) & 0 \end{pmatrix}_{\alpha\beta},$$

where the $S^{p_1p_2}$ is defined as:

$$\begin{aligned} S_1^{p_1p_2} &= \tilde{\nu} I^{p_1p_2}(1 + i\tilde{\nu}, 1 - i\tilde{\nu}, 1 + i\tilde{\nu}), \\ S_2^{p_1p_2} &= \tilde{\nu} I^{p_1p_2}(i\tilde{\nu}, -i\tilde{\nu}, i\tilde{\nu}), \\ S_3^{p_1p_2} &= \tilde{\nu} J^{p_2p_1}(i\tilde{\nu}, -i\tilde{\nu}, i\tilde{\nu}), \\ S_4^{p_1p_2} &= \tilde{\nu} J^{p_2p_1}(1 + i\tilde{\nu}, 1 - i\tilde{\nu}, 1 + i\tilde{\nu}), \end{aligned}$$

and function $I^{p_1p_2}(x, y, z)$ and $J^{p_1p_2}(x, y, z)$ is:

$$\begin{aligned} I^{p_1p_2}(x, y, z) &= i2^{-2-p_1-p_2} e^{-\frac{i\pi}{2}(p_1+p_2)} \pi \text{csch}(2\pi\tilde{\nu}) \Gamma(2 + p_1 + p_2) \\ &\times \left[\frac{e^{-\pi\tilde{\nu}}}{\Gamma(x)} \Gamma(1 + p_1 - i\tilde{\nu}) \Gamma(2 + p_1 + p_2 - i2\tilde{\nu}) \right. \\ &\times {}_4\tilde{F}_3 \left(\begin{matrix} 2 + p_1 + p_2, & 1 + p_1 - i\tilde{\nu}, & y, & 2 + p_1 + p_2 - i2\tilde{\nu} \\ 2 + p_1 - i\tilde{\nu}, & 3 + p_1 + p_2 - z, & 1 - i2\tilde{\nu} \end{matrix} \middle| -1 \right) \\ &\left. - (\tilde{\nu} \rightarrow -\tilde{\nu}) \right] \end{aligned} \quad (30)$$

$$\begin{aligned} J^{p_1p_2}(x, y, z) &= i2^{-2-p_1-p_2} e^{\frac{i\pi}{2}(p_1+p_2)} \pi \text{csch}(2\pi\tilde{\nu}) \Gamma(2 + p_1 + p_2) \\ &\times \left[\frac{e^{\pi\tilde{\nu}}}{\Gamma(x)} \Gamma(1 + p_1 - i\tilde{\nu}) \Gamma(2 + p_1 + p_2 - i2\tilde{\nu}) \right. \\ &\times {}_4\tilde{F}_3 \left(\begin{matrix} 2 + p_1 + p_2, & 1 + p_1 - i\tilde{\nu}, & y, & 2 + p_1 + p_2 - i2\tilde{\nu} \\ 2 + p_1 - i\tilde{\nu}, & 3 + p_1 + p_2 - z, & 1 - i2\tilde{\nu} \end{matrix} \middle| -1 \right) \\ &\left. - (\tilde{\nu} \rightarrow -\tilde{\nu}) \right] \end{aligned} \quad (31)$$

Also for the right subgraph:

$$\begin{aligned} \mathcal{T}_{f\gamma\rho}^{(R)} &= \left(\frac{y^t}{\sqrt{2}} \right)^2 \frac{H^2}{4k_3^3 k_4^3} \frac{1}{k_3^{if\tilde{\nu}} k_4^{if\tilde{\nu}}} \\ &\times \left[\hat{I}_{ff\gamma\rho}^{-1+if\tilde{\nu}, -1+if\tilde{\nu}} - if \hat{I}_{ff\gamma\rho}^{if\tilde{\nu}, -1+if\tilde{\nu}} - if \hat{I}_{ff\gamma\rho}^{-1+if\tilde{\nu}, if\tilde{\nu}} - \hat{I}_{ff\gamma\rho}^{if\tilde{\nu}, if\tilde{\nu}} \right] \end{aligned} \quad (32)$$

We set $\mathbf{k}_1 = -\mathbf{k}_2 = (0, 0, k_1)$, and $\mathbf{k}_3 = -\mathbf{k}_4 = (k_3 \sin \theta, 0, k_3 \cos \theta)$, so there is an additional rotation matrix will act on $\mathcal{T}^{(R)}$:

$$\mathcal{T}_{f\gamma\rho}^{(R)} \Rightarrow U_\gamma{}^\lambda U_\rho{}^\sigma \mathcal{T}_{f\lambda\sigma}^{(R)} \quad (33)$$

where U is the rotation matrix:

$$U(\theta, 0) = \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix} \quad (34)$$

So the total trispectrum of by $4\delta h\bar{\psi}\bar{\psi}$ diagram:

$$\langle \delta h \rangle^{40} = \sum_{f=\pm} [\mathcal{B}_{f,\beta}^{\gamma\alpha}] [\mathcal{T}_f^{(L)}] \epsilon^{\beta\kappa} [U_\gamma^\lambda \mathcal{T}_f^{(R)\sigma} (U^{-1})_\sigma^\delta] \epsilon^{\delta\rho} \quad (35)$$

By simplifying the spinor structure and some numerical check, we find that the final result:

$$\langle \delta h \rangle^{40} = \left(\frac{y^t}{\sqrt{2}} \right)^4 \frac{H^4 k_s^3}{k_1^3 k_2^3 k_3^3 k_4^3} \left[C^{40}(\tilde{\nu}) \left(\frac{k_s^2}{4k_{12}k_{34}} \right)^{2i\tilde{\nu}} (1 + \cos \theta) + \text{c.c.} \right] \quad (36)$$

where the magnitude:

$$C^{40}(\tilde{\nu}) = \mathcal{B}_+ \cdot 16^{-1+2i\tilde{\nu}} \cdot 2 \left[\hat{I}_{++12}^{-1+i\tilde{\nu}, -1+i\tilde{\nu}} - i \hat{I}_{++12}^{i\tilde{\nu}, -1+i\tilde{\nu}} - i \hat{I}_{++12}^{-1+i\tilde{\nu}, i\tilde{\nu}} - \hat{I}_{++12}^{i\tilde{\nu}, i\tilde{\nu}} \right]^2 \quad (37)$$

And a numerical result:

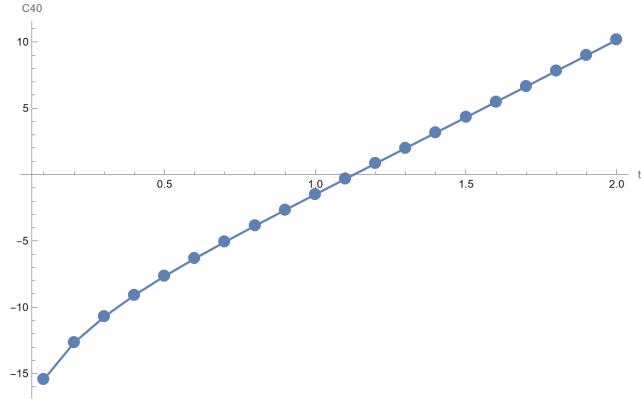


Figure 1: $|C^{40}(\tilde{\nu})|$

Also for the contribution of $4\delta h\bar{\psi}^\dagger\bar{\psi}^\dagger$ diagram, we just replace the integral seed by corresponding seed:

$$\langle \delta h \rangle^{04} = \langle \delta h \rangle^{40}|_{\hat{I} \rightarrow \bar{I}} = \sum_{f=\pm} [\mathcal{B}_{f,\beta}^{\gamma\alpha}]|_{\alpha \rightarrow \dot{\alpha}} [\mathcal{T}_f^{(L)\beta}]|_{\hat{I} \rightarrow \bar{I}} [U_\gamma^\lambda \mathcal{T}_f^{(R)\sigma} (U^{-1})_\sigma^\delta]|_{\hat{I} \rightarrow \bar{I}} \quad (38)$$

$$= \left(\frac{y^t}{\sqrt{2}} \right)^4 \frac{H^4 k_s^3}{k_1^3 k_2^3 k_3^3 k_4^3} \left[C^{04}(\tilde{\nu}) \left(\frac{k_s^2}{4k_{12}k_{34}} \right)^{2i\tilde{\nu}} + \text{c.c.} \right] \quad (39)$$

where the magnitude:

$$C^{04}(\tilde{\nu}) = [C^{40}(\tilde{\nu})]^* \quad (40)$$

1.2 $2 - \delta h \bar{\psi} \bar{\psi}$ & $2 - \delta h \bar{\psi}^\dagger \bar{\psi}^\dagger$ Diagram

There only only 1 type of graph will not vanish:

$$\begin{aligned}
\langle \delta h \rangle^{40} &= 2 \cdot \left(\frac{y^t}{\sqrt{2}} \right)^4 \int \frac{d\tau_1}{(-H\tau_1)} \frac{d\tau_2}{(-H\tau_2)} \frac{d\tau_3}{(-H\tau_3)} \frac{d\tau_4}{(-H\tau_4)} \int \frac{d^3\mathbf{q}}{(2\pi)^3} \sum_{a,b,c,d=\pm} (abcd) \\
&\times G_a(k_1, \tau_1) G_b(k_2, \tau_2) G_c(k_3, \tau_3) G_d(k_4, \tau_4) \\
&\times D_{ab\alpha\beta'}(\mathbf{q} + \mathbf{k}_1, \tau_1, \tau_2) \epsilon^{\dot{\beta}'\dot{\beta}} \check{D}_{bc\dot{\beta}\dot{\gamma}'}(\mathbf{q} + \mathbf{k}_s, \tau_2, \tau_3) \epsilon^{\dot{\gamma}'\dot{\gamma}} \\
&\times D_{cd\delta'\dot{\gamma}}(\mathbf{q} + \mathbf{k}_3, \tau_3, \tau_4) \epsilon^{\delta'\delta} \hat{D}_{da\delta\alpha'}(\mathbf{q}, \tau_4, \tau_1) \epsilon^{\alpha'\alpha}
\end{aligned} \tag{41}$$

So the Bubble Contribution:

$$\mathcal{B}_{f,\dot{\beta}}^{\dot{\gamma}\alpha} = \mathcal{B}_f \cdot k_s^{3+4if\tilde{\nu}} \cdot \delta_{\dot{\beta}}^{\dot{\gamma}} \delta_{\delta}^{\alpha} \tag{42}$$

where scalar function $\mathcal{B}_f$ is identical to last subsection.

And for left subgraph, we just replace the integral seed and deal with the spinor index carefully:

$$\begin{aligned}
\mathcal{T}_{f\alpha\dot{\beta}'}^{(L)} &= \left(\frac{y^t}{\sqrt{2}} \right)^2 \frac{H^2}{4k_1^3 k_2^3} \frac{1}{k_1^{if\tilde{\nu}} k_2^{if\tilde{\nu}}} \\
&\times \left[I_{ff\alpha\dot{\beta}'}^{-1+if\tilde{\nu}, -1+if\tilde{\nu}} - if I_{ff\alpha\dot{\beta}'}^{if\tilde{\nu}, -1+if\tilde{\nu}} - if I_{ff\alpha\dot{\beta}'}^{-1+if\tilde{\nu}, if\tilde{\nu}} - I_{ff\alpha\dot{\beta}'}^{if\tilde{\nu}, if\tilde{\nu}} \right]
\end{aligned} \tag{43}$$

where the integral seed:

$$\mathcal{I}_{\oplus\oplus\alpha\dot{\beta}}^{p_1 p_2} = \begin{pmatrix} A_1^{p_1 p_2} - B_2^{p_1 p_2} & 0 \\ 0 & A_2^{p_1 p_2} - B_1^{p_1 p_2} \end{pmatrix}_{\alpha\dot{\beta}}, \tag{44}$$

$$\mathcal{I}_{\ominus\ominus\alpha\dot{\beta}}^{p_1 p_2} = \left(\mathcal{I}_{\oplus\oplus\beta\dot{\alpha}}^{p_2^* p_1^*} \right)^\dagger = \begin{pmatrix} (A_1^{p_2^* p_1^*} - B_2^{p_2^* p_1^*})^* & 0 \\ 0 & (A_2^{p_2^* p_1^*} - B_1^{p_2^* p_1^*})^* \end{pmatrix}_{\alpha\dot{\beta}}, \tag{45}$$

and the components A and B are:

$$A_1^{p_1 p_2} = \tilde{\nu}^2 I^{p_1 p_2} (1 + i\tilde{\nu}, 1 - i\tilde{\nu}, i\tilde{\nu}), \tag{46}$$

$$A_2^{p_1 p_2} = I^{p_1 p_2} (i\tilde{\nu}, -i\tilde{\nu}, 1 + i\tilde{\nu}), \tag{47}$$

$$B_1^{p_1 p_2} = I^{p_2 p_1} (i\tilde{\nu}, -i\tilde{\nu}, 1 + i\tilde{\nu}), \tag{48}$$

$$B_2^{p_1 p_2} = \tilde{\nu}^2 I^{p_2 p_1} (1 + i\tilde{\nu}, 1 - i\tilde{\nu}, i\tilde{\nu}). \tag{49}$$

And the right subgraph:

$$\begin{aligned}
\mathcal{T}_{f\delta'\dot{\gamma}}^{(R)} &= \left(\frac{y^t}{\sqrt{2}} \right)^2 \frac{H^2}{4k_3^3 k_4^3} \frac{1}{k_3^{if\tilde{\nu}} k_4^{if\tilde{\nu}}} \\
&\times \left[I_{ff\delta'\dot{\gamma}}^{-1+if\tilde{\nu}, -1+if\tilde{\nu}} - if I_{ff\delta'\dot{\gamma}}^{if\tilde{\nu}, -1+if\tilde{\nu}} - if I_{ff\delta'\dot{\gamma}}^{-1+if\tilde{\nu}, if\tilde{\nu}} - i I_{ff\delta'\dot{\gamma}}^{if\tilde{\nu}, if\tilde{\nu}} \right]
\end{aligned} \tag{50}$$

So the total trispectrum:

$$\langle \delta h \rangle^{22} = 2 \cdot \sum_{f=\pm} [\mathcal{B}_{f,\dot{\beta}}^{\dot{\gamma}\alpha}] [\mathcal{T}_{f\alpha\dot{\beta}'}^{(L)}] \epsilon^{\dot{\beta}'\dot{\beta}} [U_{\delta'}^{\lambda'} U_{\dot{\gamma}}^{*\sigma} \mathcal{T}_{f\lambda'\dot{\sigma}}^{(R)}] \epsilon^{\delta'\delta} \tag{51}$$

After carefully dealing with the spinor structure:

$$\langle \delta h \rangle^{22} = 2 \cdot \left(\frac{y^t}{\sqrt{2}} \right)^4 \frac{H^4 k_s^3}{k_1^3 k_2^3 k_3^3 k_4^3} \left[C_+^{22}(\tilde{\nu}) \left(\frac{k_s^2}{4k_{12}k_{34}} \right)^{2i\tilde{\nu}} + C_-^{22}(\tilde{\nu}) \left(\frac{k_s^2}{4k_{12}k_{34}} \right)^{-2i\tilde{\nu}} \right] \quad (52)$$

where the magnitude:

$$\begin{aligned} C_+^{22}(\tilde{\nu}) &= -\mathcal{B}_+ \cdot 16^{-1+2i\tilde{\nu}} \cdot 2 \left[\frac{(I_{++11} + I_{++22})^2}{2} - \frac{(I_{++11} - I_{++22})^2}{2} \cos \theta \right] \\ C_-^{22}(\tilde{\nu}) &= -\mathcal{B}_- \cdot 16^{-1-2i\tilde{\nu}} \cdot 2 \left[\frac{(I_{--11} + I_{--22})^2}{2} - \frac{(I_{--11} - I_{--22})^2}{2} \cos \theta \right] \end{aligned} \quad (53)$$

where the matrix element:

$$\begin{aligned} I_{++11} &= I_{++11}^{-1+i\tilde{\nu}, -1+i\tilde{\nu}} - iI_{++11}^{i\tilde{\nu}, -1+i\tilde{\nu}} - iI_{++11}^{-1+i\tilde{\nu}, i\tilde{\nu}} - I_{++11}^{i\tilde{\nu}, i\tilde{\nu}} \\ I_{++22} &= I_{++22}^{-1+i\tilde{\nu}, -1+i\tilde{\nu}} - iI_{++22}^{i\tilde{\nu}, -1+i\tilde{\nu}} - iI_{++22}^{-1+i\tilde{\nu}, i\tilde{\nu}} - I_{++22}^{i\tilde{\nu}, i\tilde{\nu}} \\ I_{--11} &= I_{--11}^{-1+i\tilde{\nu}, -1+i\tilde{\nu}} - iI_{--11}^{i\tilde{\nu}, -1+i\tilde{\nu}} - iI_{--11}^{-1+i\tilde{\nu}, i\tilde{\nu}} - I_{--11}^{i\tilde{\nu}, i\tilde{\nu}} \\ I_{--22} &= I_{--22}^{-1+i\tilde{\nu}, -1+i\tilde{\nu}} - iI_{--22}^{i\tilde{\nu}, -1+i\tilde{\nu}} - iI_{--22}^{-1+i\tilde{\nu}, i\tilde{\nu}} - I_{--22}^{i\tilde{\nu}, i\tilde{\nu}} \end{aligned} \quad (54)$$

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